

Temporal chromatin acetylation dynamics at the Cxcl9–Cxcl11 locus

Genome browser tracks across mm10 chr5:92.32–92.38 Mb reveal a coordinated, transient wave of histone acetylation centered on the Cxcl9–Cxcl11 chemokine locus with maximal breadth at 6 hr and focal persistence by 24 hr (Figure 1). With fixed y-axis scaling (H3K27ac: 0–9; H2BK12ac/H4K5ac: 0–3) to enable direct temporal comparisons, all three marks are near baseline at 0 hr, undergo robust induction at 6 hr, and contract to discrete peaks at 24 hr. Among the marks, H3K27ac exhibits the broadest enrichment, forming a locus-spanning domain across the Cxcl10–Cxcl11 interval and extending toward Gm2673 at 6 hr; H2BK12ac shows a similar but narrower pattern, and H4K5ac is more restricted in breadth. By 24 hr, signals diminish globally yet persist as punctate peaks over a subset of sites, with prominent retention within the Cxcl10–Cxcl11 interval and scattered positions across the region.

These dynamics are consistent with rapid, stimulus-coupled activation of regulatory chromatin across the locus—most conspicuously captured by H3K27ac, a mark associated with active promoters and enhancers—followed by consolidation to a limited set of elements that likely sustain ongoing regulatory output at later time points (Bellmunt et al., 2026). The broad 6 hr H3K27ac domain suggests widespread enhancer engagement across the Cxcl10–Cxcl11 interval, whereas the 24 hr pattern indicates selective maintenance of promoter-proximal and/or key enhancer sites, particularly at Cxcl10/Cxcl11. The concordant, albeit more spatially confined, induction of H2BK12ac and H4K5ac supports coordinated promoter and gene-proximal activation with temporal narrowing by 24 hr. Together, these fixed-scale profiles indicate a temporally resolved program in which early

locus-wide acetylation transitions to focal, sustained activation within the *Cxcl10*–*Cxcl11* regulatory landscape.

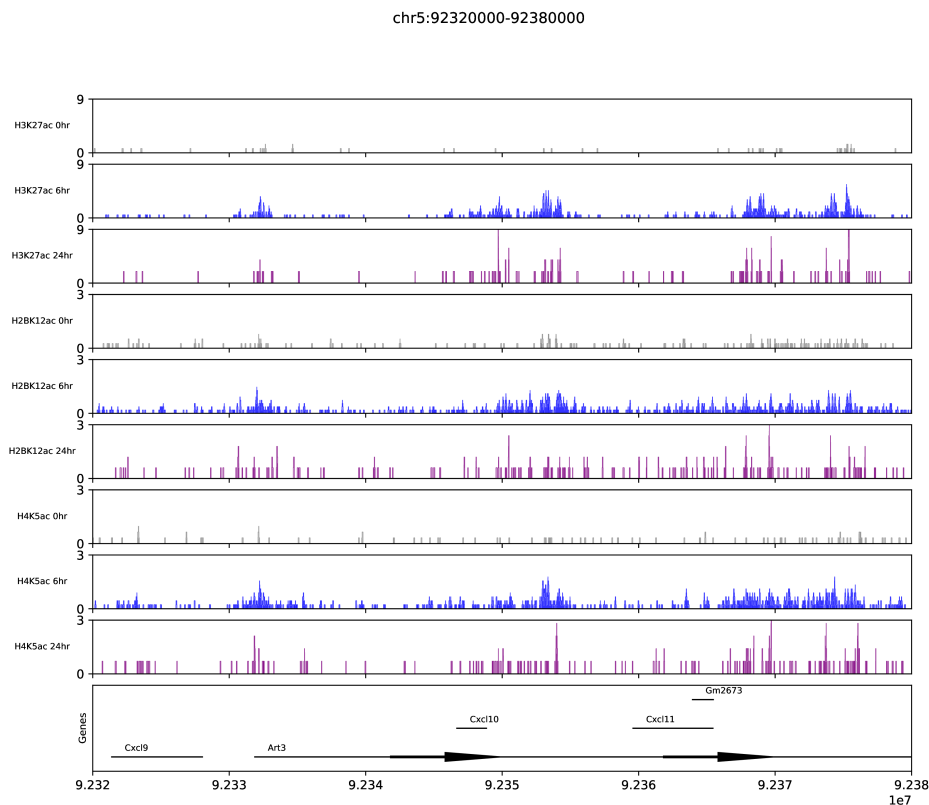


Figure 1. Temporal acetylation dynamics at the *Cxcl9*–*Cxcl11* locus. Genome coverage tracks for *H3K27ac*, *H2BK12ac*, and *H4K5ac* across mm10 chr5:92.32–92.38 Mb compare 0 hr, 6 hr, and 24 hr. Signals are low at 0 hr, rise broadly at 6 hr across the *Cxcl10*/*Cxcl11* region, and persist as narrower peaks at selected sites by 24 hr.

References

1. Bellmunt, J., Xie, Y., Juanpere, N., Munoz, M. G., Kukreja, S., Veiga, S. L., Li, R., Qiu, X., Jiang, Y., Font-Tello, A., Duarte, M. N., Epstein, I., Hernández-Llodrá, S., Lorenzo, M., Menendez, S., Choueiri, T., Brown, M., Long, H. W., & Cejas, P. (2026). Epigenetic subtypes of high-grade T1 bladder cancer reveal intra-tumor heterogeneity and distinct interactions with tumor microenvironment. *Theranostics*, 16(12), 6892–6910. <https://doi.org/10.7150/thno.129729>
2. Horst, I., Offermann, S., Dreesen, B., Niessen, M., & Peterhansel, C. (2009). Core promoter acetylation is not required for high transcription from the phosphoenolpyruvate carboxylase promoter in maize. *Epigenetics & Chromatin*, 2(1). <https://doi.org/10.1186/1756-8935-2-17>